

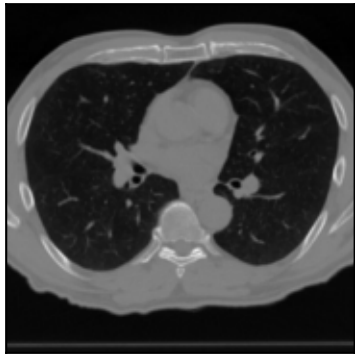
DVF QA pipeline report

DIR-Lab-case2

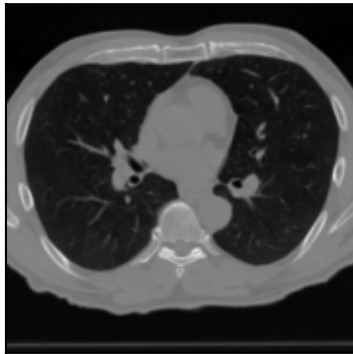
Field	Value
Case	DIR-Lab-case2
Generated	2026-05-20 09:24:14
source	Case2Pack
fps	15.0
n_cycles	4
frames_per_cycle	30
n_phases	10
ct_shape	(112, 256, 256)
spacing_xyz_mm	(1.16, 1.16, 2.5)
detector_shape	(256, 256)
pixel_spacing_mm	(1.0, 1.0)
device	cpu

4DCT phase preview

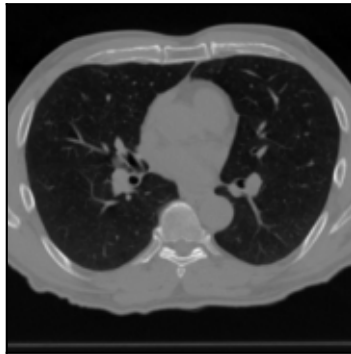
ϕ_0



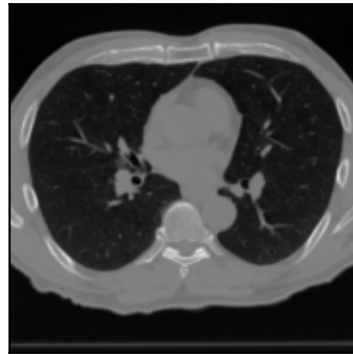
ϕ_1



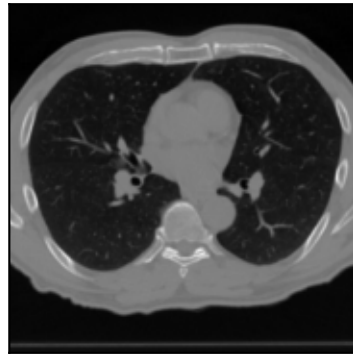
ϕ_2



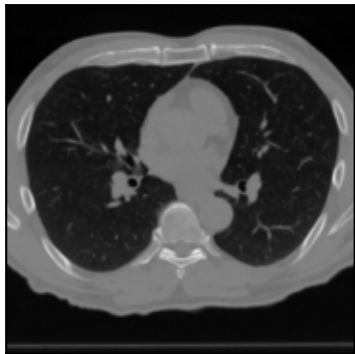
ϕ_3



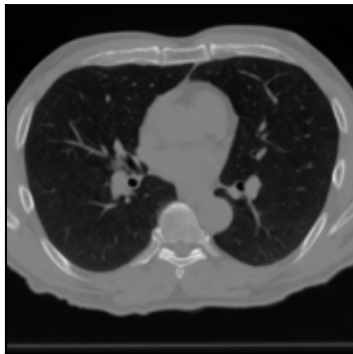
ϕ_4



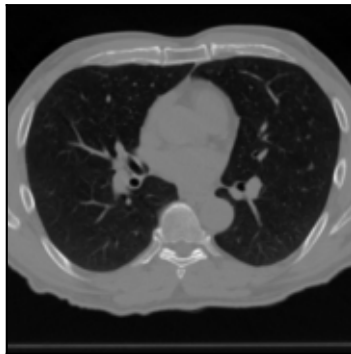
ϕ_5



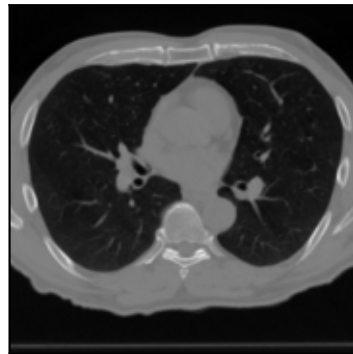
ϕ_6



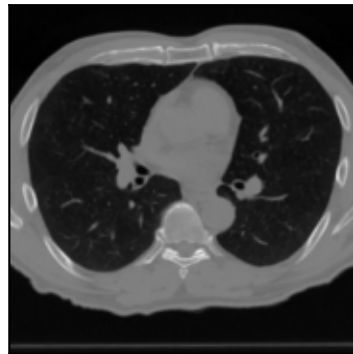
ϕ_7



ϕ_8

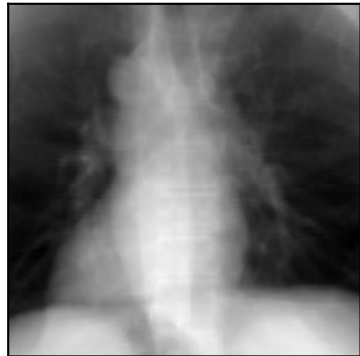


ϕ_9

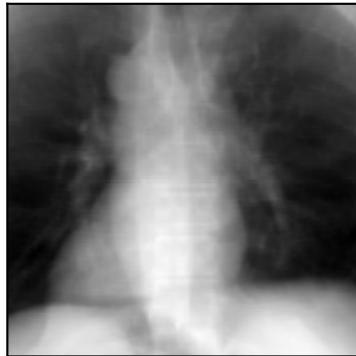


Per-phase DRRs

AP φ_0



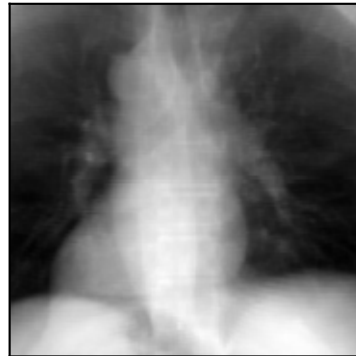
AP φ_1



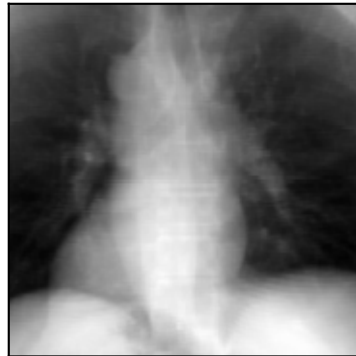
AP φ_2



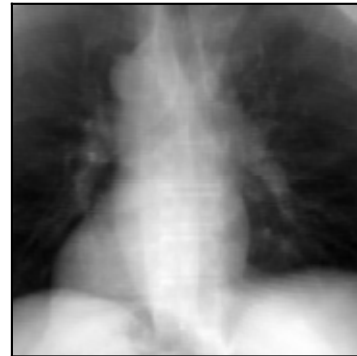
AP φ_3



AP φ_4



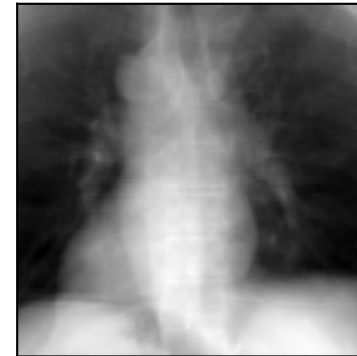
AP φ_5



AP φ_6



AP φ_7



AP φ_8



AP φ_9



Lat φ_0



Lat φ_1



Lat φ_2



Lat φ_3



Lat φ_4



Lat φ_5



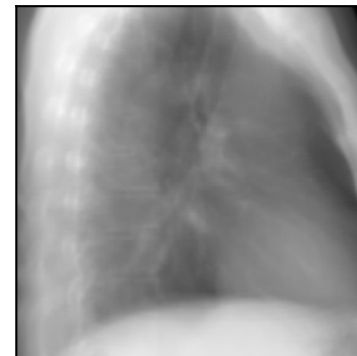
Lat φ_6



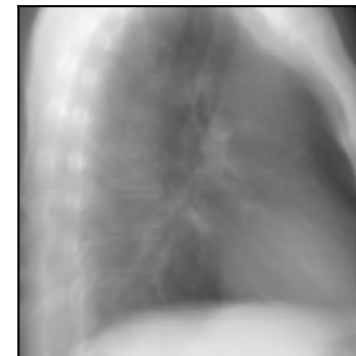
Lat φ_7



Lat φ_8



Lat φ_9



Dynamic sequence samples

AP t=0 (0.00s)



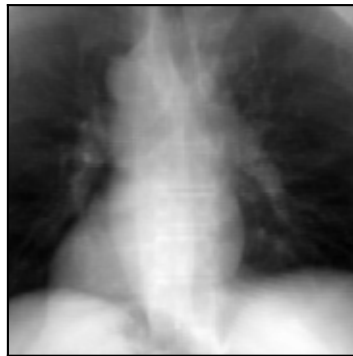
AP t=24 (1.60s)



AP t=48 (3.20s)



AP t=72 (4.80s)



AP t=96 (6.40s)



AP t=120 (8.00s)



Lat t=0



Lat t=24



Lat t=48



Lat t=72



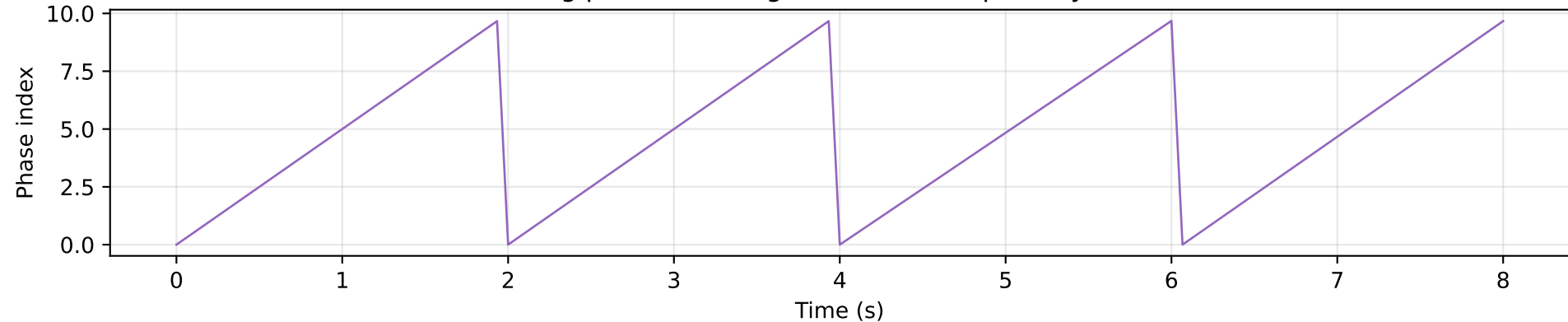
Lat t=96



Lat t=120

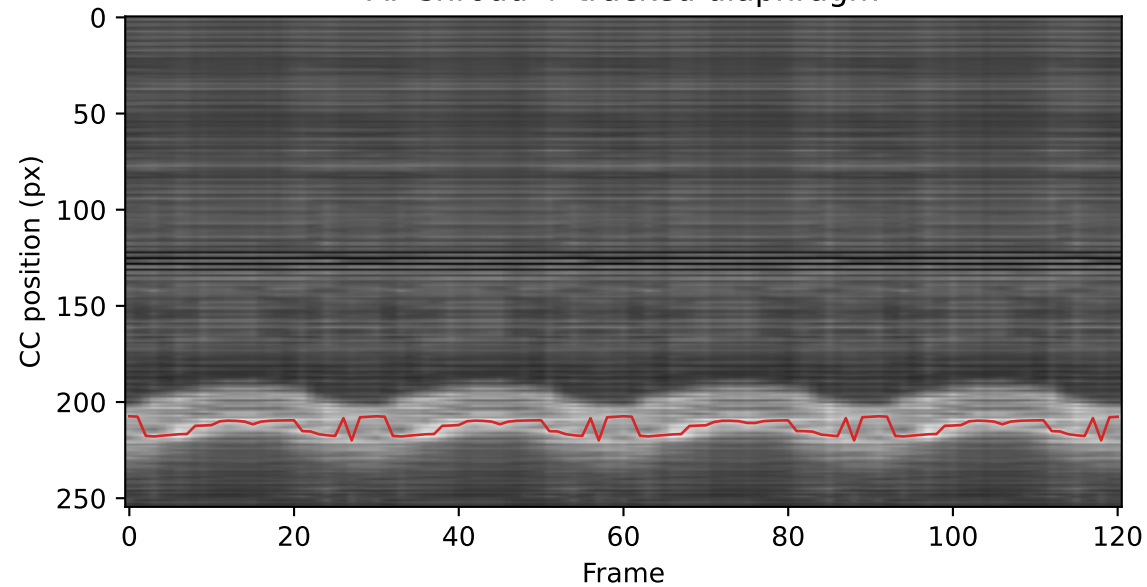


Driving phase curve
Driving phase curve (ground-truth respiratory motion)

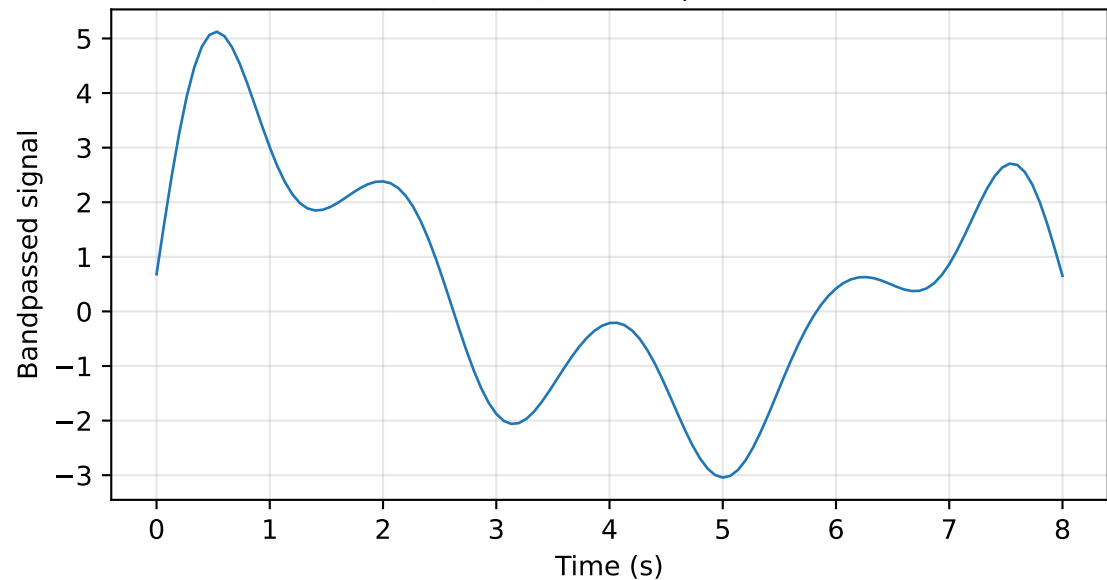


Amsterdam Shroud — AP

AP shroud + tracked diaphragm

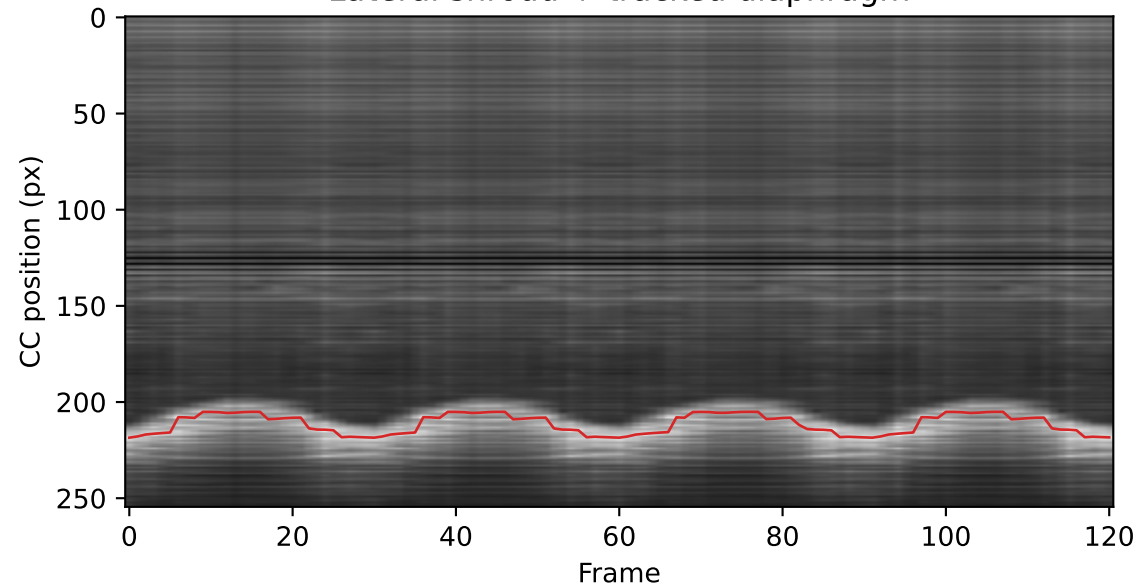


AP respiratory signal | $f=0.124$ Hz

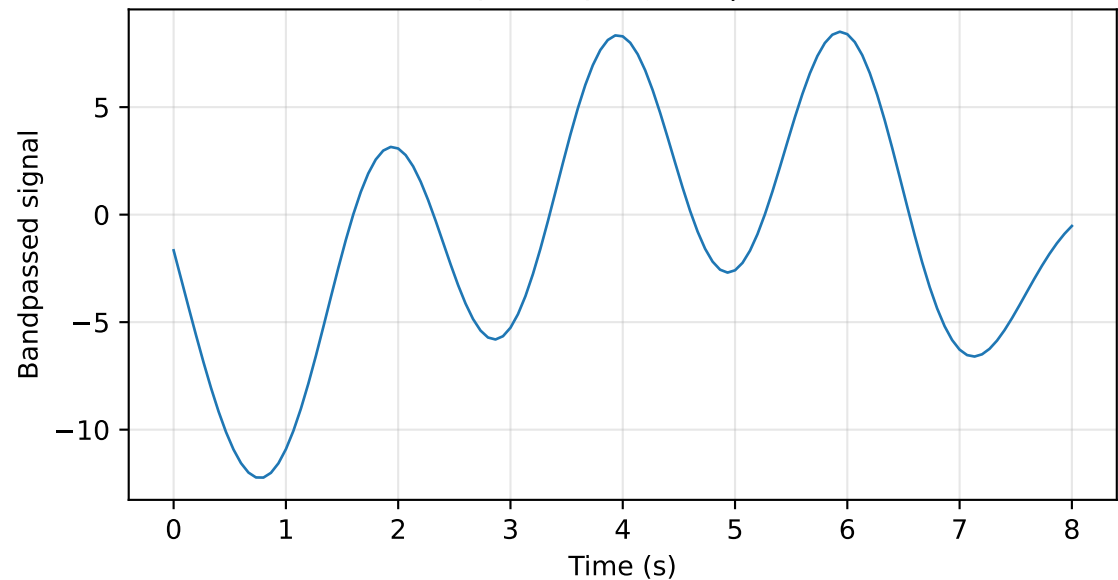


Amsterdam Shroud — Lateral

Lateral shroud + tracked diaphragm

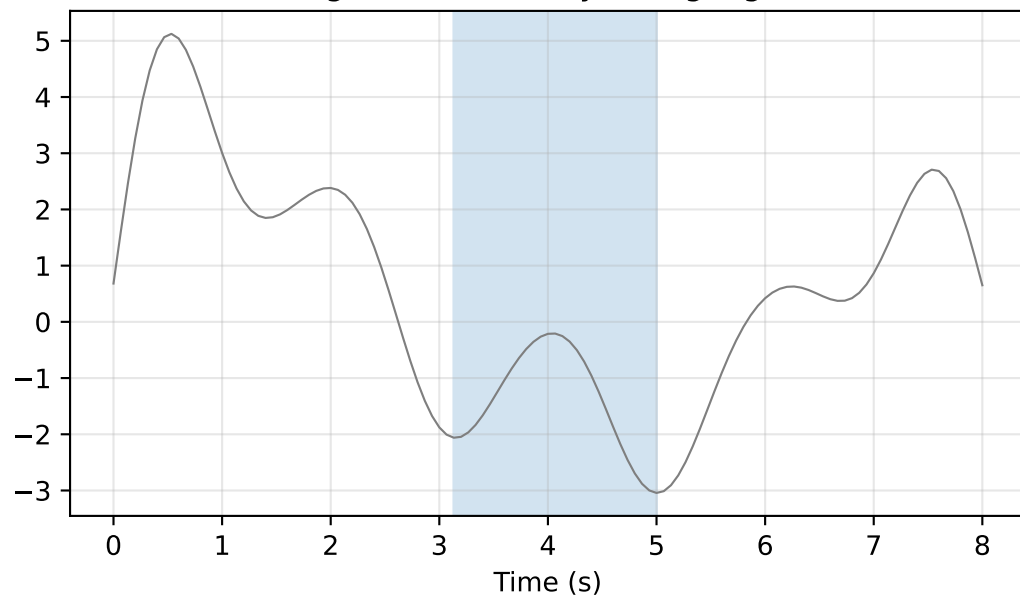


Lateral respiratory signal | $f=0.496$ Hz

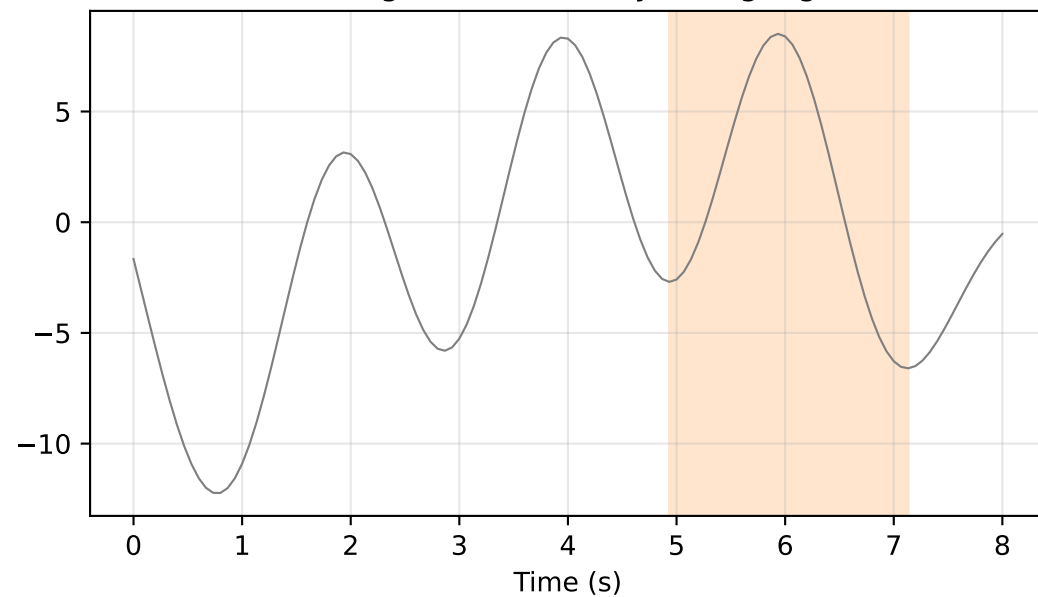


Cycle pairing

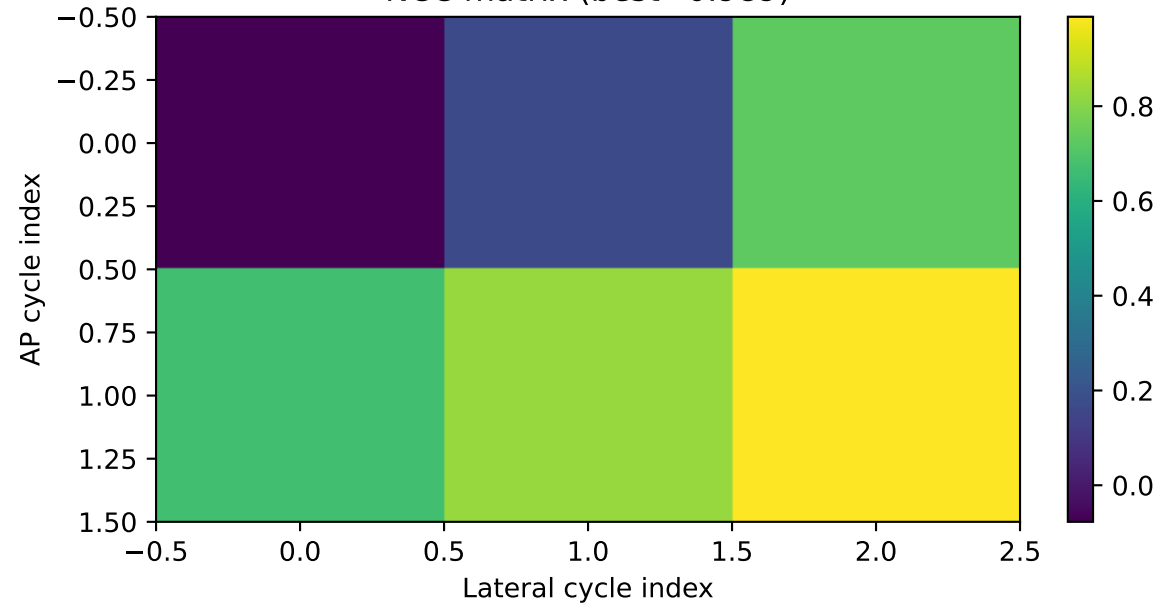
AP signal (selected cycle highlighted)



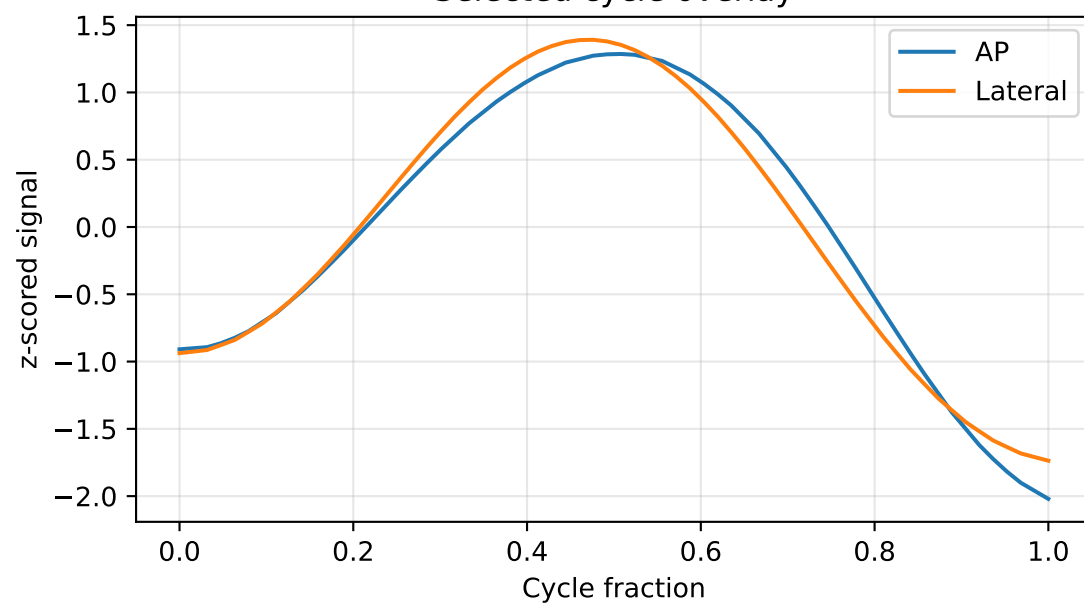
Lateral signal (selected cycle highlighted)



NCC matrix (best=0.989)



Selected cycle overlay



Metrics

Metric	Value
AP dominant frequency (Hz)	0.1240
AP period (frames)	121.00
Lateral dominant frequency (Hz)	0.4959
Lateral period (frames)	30.25
Best pair NCC	0.9886
AP cycle frames	47 - 75
Lateral cycle frames	74 - 107
min_period_s	1.0000